



GATE CS / IT 2027

Operating Systems

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

CPU Scheduling

Algorithm	Preemptive	Starvation	Key Property
FCFS	No	No	Convoy effect; simple
SJF	No	Yes	Optimal avg wait; needs future knowledge
SRTF	Yes	Yes	Preemptive SJF; optimal avg wait
Round Robin	Yes	No	Fair; time quantum critical
Priority	Both	Yes	Aging solves starvation
MLFQ	Yes	Possible	Approximates SRTF adaptively

Metrics

- Turnaround Time = Completion - Arrival
- Waiting Time = Turnaround - Burst
- Response Time = First CPU - Arrival
- Throughput = processes completed per unit time

Process Synchronization

Critical Section & Semaphores

- 3 requirements: Mutual Exclusion, Progress, Bounded Waiting
- Semaphore wait(): $S > 0$ then $S--$; else block
- Semaphore signal(): wake blocked; or $S++$
- Binary semaphore = mutex (0 or 1)
- Producer-Consumer: two semaphores (empty, full) + mutex
- Deadlock conditions: Mutual Exclusion, Hold&Wait, No Preemption, Circular Wait

Deadlock

- Banker's Algorithm: safe state check for avoidance
- Prevention: deny one of 4 deadlock conditions
- Detection: resource allocation graph or wait-for graph
- Recovery: preempt resources, rollback, kill process

Memory Management

Paging

- Physical addr = frame_number * page_size + offset
- TLB speeds up virtual-to-physical translation
- 2-level/multi-level page table for large address space
- Inverted page table: one entry per frame

Page Replacement

- FIFO: replace oldest (Belady anomaly possible)
- Optimal: replace page used farthest in future
- LRU: replace least recently used (best practical)
- Clock (Second Chance): efficient LRU approximation

File Systems

Allocation & Disk

- Contiguous: fast sequential access, external fragmentation
- Linked: no external fragmentation, slow random access
- Indexed (inode): direct + indirect + double-indirect blocks
- Disk: FCFS simple, SSTF greedy, SCAN/CSCAN elevator